I. Overview

Besides disrupting global trade markets[1] and deteriorating the quality of the United States' trade partnerships, Trump's 2025 tariffs are expected to directly affect the welfare of US citizens. Tariffs, a measure typically intended to raise revenue for the implementing country's government and encourage domestic production, have seen new appraisal in the aftermath of drastic tariff increases imposed by President Trump in 2025. By definition[2], tariffs are applied to the base price of exported items, which is the price of the exported goods in their native currency and additional shipment costs borne by the exporter. Tariffs increase the total cost of the exported goods, which is directly reflected in higher prices paid by importing buyers. To compensate for the increase in import costs, the importing buyers (who are domestic suppliers of goods) raise market prices (what consumers pay) by the level of price change caused by the tariffs. According to basic supply and demand theory, increases in consumer prices result in fewer individuals who are willing to buy the goods at the higher price. When there is less demand for goods, the supply of goods produced also decreases. This has the compound effect of threatening the stability of good supplies, jobs, and consumer prices.

In addition to import price increases, tariffs also have broader impacts on the global trade market. The global trade market is characterized by trade agreements, in which all involved trade partners exchange goods according to agreed-upon regulations. Trump's

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[1] J.P. Morgan Global Research (2025, December 5). *US tariffs: What's the impact on global trade and the economy?* Jpmorgan.com. Retrieved December 12, 2025, from https://www.jpmorgan.com/insights/global-research/current-events/us-tariffs

[2] TaxEDU (n.d.). *Tariff*. Taxfoundation.org. https://taxfoundation.org/taxedu/glossary/tariffs/

tariffs represent decisions made unilaterally by the United States, which violate the implicit and explicit (shockingly, Trump's tariffs are illegal, yet persist) global trade agreements. In response, countries subject to the sudden tariffs have reacted in three main ways[3]: by imposing equal retaliatory tariffs, altering domestic policy to support industry producers, and/or opting for new trading partners. The impacts of this are seen tangibly through a drastic reduction in the quantity of imports (as exporting countries are less willing to trade with the US, and demand decreases with higher prices). Or so I expected. During my exploratory data analysis, however, I discovered that contrary to my expectations, the United States' global import and export quantities have been fairly consistent with pre-tariff announcement trends[4]. There was a spike in imports at the end of 2024, which I believe can be attributed to expectations of higher import prices under the assumption that Trump's tariffs would go into effect at the beginning of his presidency in 2025. Other than that, exports have been rising gradually, and imports appear to have stabilized following the spike.

Given the complex nature of tariffs and their interrelation with the domestic economy and global market dynamics, I perform a relatively simple investigation. For this project, I only attempt to offer a partial analysis of the impacts of the 2025 tariffs. By focusing on the effects on producer and consumer prices in the US, I hope that with future analysis, I could then attempt to estimate higher-order effects, such as changes to domestic industry prices and consumption behaviors.

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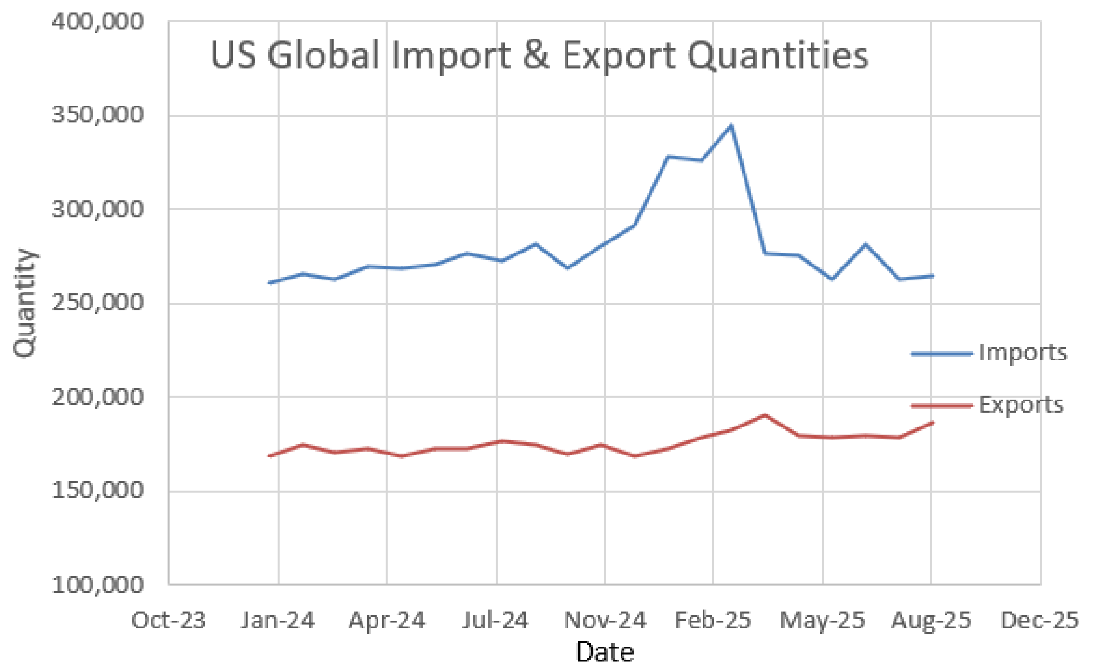
[3] Rotunno, L., & Ruta, M. (2025). Trade Partners' Responses to US Tariffs. *International Monetary Fund Working Paper*.

[4] United States Census Bureau, [DATASET] Trade in Goods with World, Seasonally Adjusted, 2024-2025. https://www.census.gov/foreign-trade/balance/c0004.html

Figure 1



II. Related Work

In 2021, Cavallo et al.[5] published an analysis of the effects of trade policies implemented during Trump's first term on import, export, and consumer prices. They used tariff and exchange rate pass-through models to estimate the first-order effect of tariffs on domestic prices in the US. Their model uses the approach of a full pass-through, meaning that tariff costs are fully reflected in consumer prices. A different paper by Amiti et al.[6] similarly explores the impact of Trump's 2018 tariffs on domestic prices and citizens' welfare, assuming a full pass-through of tariffs on consumer prices. Both papers claim that

[5] Cavallo, Alberto, Gita Gopinath, Brent Neiman, and Jenny Tang. 2021. "Tariff Pass-Through at the Border and at the Store: Evidence from US Trade Policy." *American Economic Review: Insights* 3 (1): 19–34.DOI: 10.1257/aeri.20190536

[6] Amiti, Mary, Stephen J. Redding, and David E. Weinstein. 2019. "The Impact of the 2018 Tariffs on Prices and Welfare." *Journal of Economic Perspectives* 33 (4): 187–210.DOI: 10.1257/jep.33.4.187

the full pass-through is a conventional economic approach for predicting the impact of taxes and tariffs on consumer prices; therefore, I decided to use this same technique. The basic equation for the full pass-through model is:

$$\mathbf{Ppost} = (1+\mathbf{t})\ \mathbf{Ppre}$$

Where **Ppost** is the import price after tariffs are applied, **t** is the applied tariff rate, and **Ppre** is the import price before tariffs. From this simple model, we can see that the estimated price change resulting from tariff increases is simply the rate of the tariff. This is a simplification of Cavallo et al.'s model, which also controlled for differences in the effects to particular industry sectors, but is a close representation of Amiti et al.'s. For the sake of simplicity, I followed the work of Amitit et al. as a guide for my general analysis. My three main areas of interest are impacts on import quantities, exporters, and domestic prices.

III. Data and Modeling

To answer this question, I began by collecting data on newly enforced tariffs. Since tariff regulations have progressively changed since the beginning of 2025, with up to several new tariff announcements made within a single month[7], I simplified the data collection process by using aggregate statistics. The two main sources I examined to extract data about Trump's Tariffs are the Atlantic Council's Trump Tariff Tracker website[8] and Reed Smith's

[7] Urban Institute & Brookings Institution Tax Policy Center (2025, December 9). *Tracking the Trump Tariffs*. Taxpolicycenter.org. Retrieved December 12, 2025, from https://taxpolicycenter.org/features/tracking-trump-tariffs
[8] Atlantic Council (2025, December 5). *Trump Tariff Tracker*. Atlanticcouncil.org. Retrieved December 3, 2025, from https://www.atlanticcouncil.org/programs/geoeconomics-center/trump-tariff-tracker/

Trump 2.0 Tariff Tracker[9]. There are a few key factors to consider when analyzing the 2025 tariffs. One is the nature of the tariffs' rates. The majority of the tariffs are *ad valorem*, meaning that they are a fixed percent of the import's value. For example, if 2 million dollars' worth of rice is imported from Senegal with an ad valorem tariff of 15%, the import price is effectively raised to 2,000,000*1.15*currency_conversion_rate. We multiply by the currency conversion rate since the tariffs are applied to the base value of incoming exports, which then needs to be converted to the US dollar value for consistency with domestic prices. The other kind of tariff rate is specific—a fixed cost per specified unit of the imported quantity. For example, if 100,000 gallons of oil are imported from Brazil with a specific tariff of 50 cents per gallon, the surcharge on the import price is 0.5 x 100,000 = 50,000 USD. For these, the only currency conversion required is to the base value of the import quantity. The tariff rates vary by type (ad valorem or specific), value, country, and industry, which complicates calculating their impact on prices. Fortunately, the Atlantic Council's Trump Tariff Tracker also contains estimates for the "additional duty" (tariff) rate applied to imports from each country. The "additional duty" represents the average increase in the duty rate on imported products from each country. For my analysis, I used these additional duty rates as the estimated percent increase in import prices, while accounting for currency conversion rates.

Another important factor for contemplation is the fact that the tariffs were implemented on different dates throughout the year. Looking through the specific tariffs and their implementation dates, I identified four main time periods. These are from January to the end of March, April to June, July to September, and October to early December. For the

[9] Lowell, M., Heeren, P., Angotti, J., Rodriguez-Johnson, L., Lowell, K., & Yaeger, E. (2025, December 9). *Trump Tariff Tracket*. Tradecomplianceresourcehub.com. Retrieved December 10, 2025, from https://www.tradecomplianceresourcehub.com/2025/12/09/trump-2-0-tariff-tracker/#country

sake of simplicity, I approximated the implementation of tariffs in gradual, quarterly waves. I will illustrate this with an example. The additional duty rate reported for Canada is 35%. For the time period from January to March, I estimate the additional duty rate on Canada to be one-fourth of 35% (8.75%); from April to June, it is one-half of 35% (17.5%); from July to September, it is 26.25%; and from October to December the additional duty realizes its full rate of 35%.

I constructed my dataset from scratch by first listing every trade partner (country) that has had new tariffs levied against it in 2025. I added a column containing the additional duty rate for each trade partner, using the estimates from the Atlantic Council's website. I added 4 columns for the currency conversion rate corresponding to each of the quarterly time periods. I also calculated an "importance" value for each country to signify what percent of total US imports are goods from that country. I did this using data on the annual customs value of total imports by country and total imports from all countries from the United States International Trade Commission's (USITC) website[10]. The customs value is the monetary worth of imported goods, in USD, before any tariffs and taxes have been accounted for.

Additionally, I added each country's export share for annual time periods from 2008 to 2022 using data from the World Integrated Trade Solution's website[11]. Export shares report the monetary value of exports sent to the US as a percentage of the exporting

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[10] United States International Trade Commission, [DATASET] Imports for Consumption: Customs Value, 2015-2025. https://dataweb.usitc.gov/trade/search/Import/HTS

[11] World Bank, World Integrated Trade Solution, [DATASET] United States Export Partner Share to World by Country 2007-2022. https://wits.worldbank.org/CountryProfile/en/Country/USA/StartYear/2007/EndYear/2022/TradeFlow/Export/Partner/BY-COUNTRY/Indicator/XPRT-PRTNR-SHR

country's total value of all exports. I deemed this to be important, as trade is a bilateral partnership where both countries act as agents on the supply and demand side. My previous data metrics capture the importance of imports from a given country to the US market supply, which influences domestic market prices and demand. On the other hand, export shares signify the importance of US trade sales relevant to the exporting country's total trade sales.

The first model I created was a time-series forecast to predict the average import price index for the months of 2025, which are not available in the current data, and for all future months in 2026. The import price index measures the change in the prices of imported goods over time. Changes in the import price index are representative of evolving import costs, exchange rates, tariff rates, and other economic factors related to supply and demand. The import price index intrinsically accounts for domestic (in the US) inflation by using the year 2000 as its base comparison metric. I used historical data from the Federal Reserve Bank of St. Louis[12], which reports the monthly US import price index from 1985-2025. My model choice was XGBoost, as it is commonly recommended for use with time-series regressions. Still, I looked into the reasoning behind this recommendation to confirm if it was right for me. The distinguishing benefits for XGBoost regression are that it handles continuous values well (which is the value type of the import price index) and automatically imposes a loss function, which reduces errors. Furthermore, the model is generally described as being an optimal regression model with industry-standard gradient boosting.

[12] Federal Reserve Bank of St Louis (n.d.). *Import Price Index (End Use): All Commodities*. Fred.Stlouisfed.org. https://fred.stlouisfed.org/series/IR

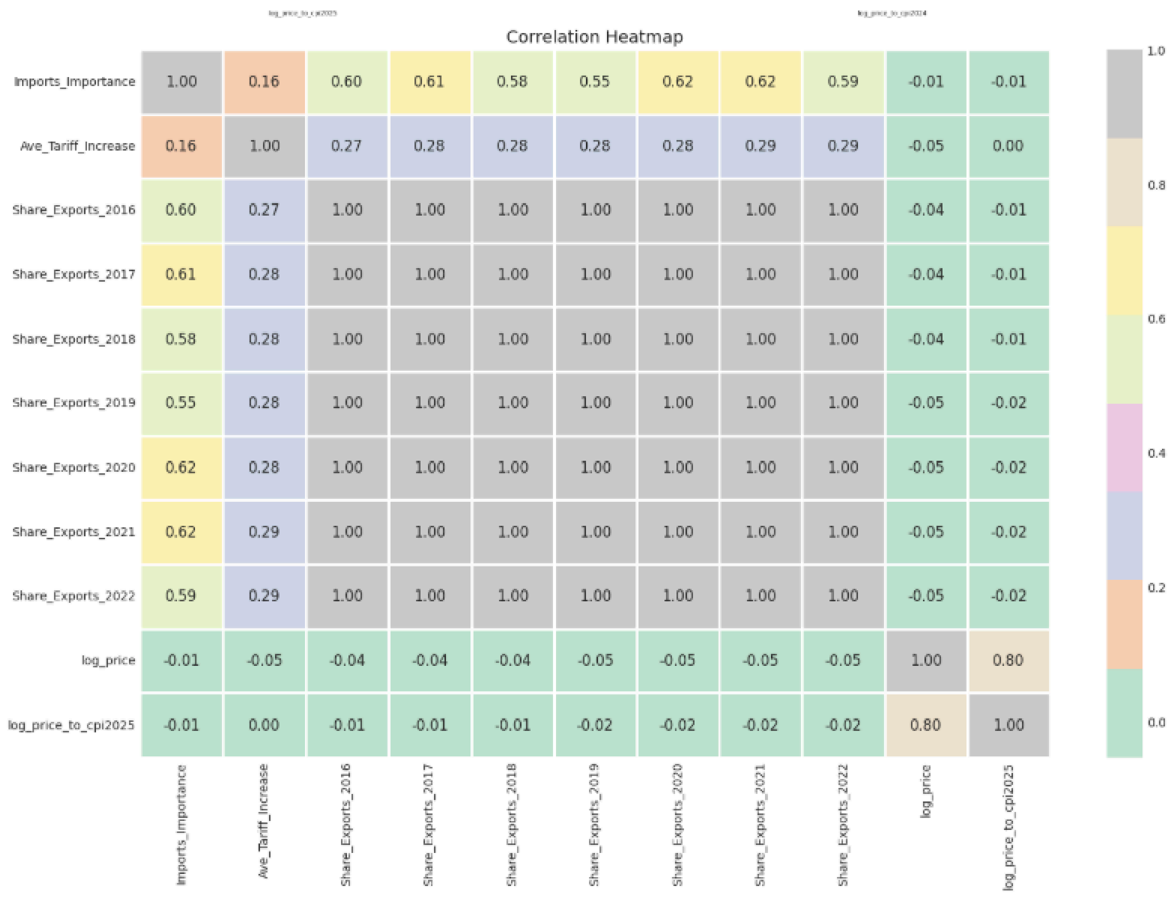
For this time series regression on historical import price indices, I had 517 observations from September 1982 up to September 2025. Given the extensive range of the time period, I assume that general trends in the change of the import price index are captured, as well as responses to underlying economic dynamics, such as financial depressions. I operated under two-types of test-train splits. For the first type, I set the test-train split date as November 1st, 2024. This date is just before the spike in imports (as seen in Figure 1 above). I was curious to see if a regression model could somehow predict the changes to import prices before the impacts of Trump's election win and the expectation of tariff increases would appear in data trends. This split resulted in only 11 test values, roughly 2% of my total observations, and clearly a violation of the typical 80/20 or even 90/10 split. However, I was curious to see how the model would perform in comparison to a more robust model with k-fold validation. For the other, more robust model, my training data consisted of the first 90% of time periods, spanning from September 1982 to May 2021. For this model, I used 5-fold cross-validation. Both test-train split types were evaluated by their performance on test data, using statistical error metrics, and by the reasonableness of their predictions for future dates under my personal judgment.

Separate from these XGBoost time-series regressions, I wanted to perform a model that included more variables. For this, I used a dataset I created with 154 observations for each country that has had an "additional duty" levied on it, as provided by the Atlantic Council's website. In addition to the additional duty rate, I included variables for the quarterly (2025) currency conversion rate, imports importance (imports to the US as a percent of global imports to the US in 2024), annual share of exports (exports to the US as a percent of total exports to all countries; from 2007-2022), and annual increases in the US consumer price index (CPI; from 2007-2025). For the change in CPI and share of exports variables, I deleted the columns for time periods that had similar correlation metrics as the columns for more recent time periods. I also deleted the columns for CPI, as the value was the same for all observations (countries) and therefore perfectly multicollinear. However, I still wanted a variable that incorporated the CPI, so I created a one for **(**log(price)/CPI_2025**)**^2. For these variables, I created the correlation matrix shown in Figure 2.

Interestingly, the correlation between log(price) and all other variables was negative (though very weak). A reminder that log(price) is my estimate for ithe ncrease in domestic prices as the percent change in the CPI from end-of-year 2024 to September of 2025, share of exports is the significance of exporting to the US relative to the exporting country's total exports, and imports importance is significance of importing country X's products relative to the US's total imports. This negative correlation implies that increases in reliance on imports from any country are weakly associated with lower rates of inflation. This is just one possible explanation, for we could just as strongly argue that there is no correlation between the two.

I used the correlation matrix to select log_price, share of exports from 2020 to 2023, imports importance, and the average tariff increase as my features. I discarded the share of exports from earlier periods as they were similar to those of later periods, but less relevant on a time-scale. For this model, I decided to use a multivariable OLS linear regression for statistical analysis. I chose this model because it handles continuous variables well and displays results with coefficients signifying the importance of each feature relative to log(price).

Figure 2



My third and last model was the simple equation for full pass-through: **Ppost** = (1+**t**) **Ppre**. For this model, I computed the Ppost with Ppre=1 for each country. I then multiplied Ppost for each country by the corresponding import importance and summed them to calculate an estimation of the percent increase in domestic prices.

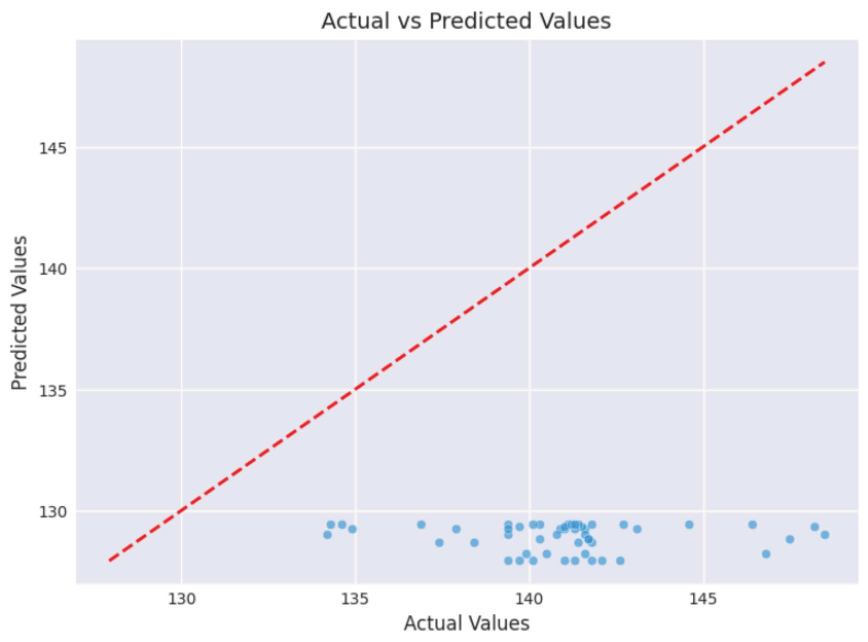
IV. Results and Conclusions

I was somewhat surprised by the results of my two XGBoost time-series regressions. The model that I split by the date just prior to Trump's election performed better than the 90-10 split with 5-fold cross-validation. My decision for a 90-10 over an 80-20 split was in efforts to improve the model's accuracy by providing it with more data. However, this did not seem to be enough. For the 5-fold, I altered the parameters several times to try to find the "best" model, which I deemed to be one with the most reasonable predictions for future, unobserved timed periods and overall higher accuracy on test-train data. The error (RMSE) for this model was similar across both testing and training data. The error was the highest during the first fold (18.2, test) and lowest in the fifth fold (4.67, train) (Figure 3). The predictions on the test data were not very accurate (Figure 4). In Figure 4, the dotted red line represents the distribution of points for exactly accurate predictions. The y-axis is the value of the model's predicted values, and the x-axis is the actual values in the dataset. The actual predictions lie in a horizontal line clustered along the bottom of the graph, completely missing the true trend line. Notably, this may be due to an error in my code or limitations of my dataset rather than weak performance from the k-fold model.

Figure 3

	train-rmse-mean	train-rmse-std	test-rmse-mean	test-rmse-std
0	17.859765	0.271444	18.204958	1.066819
1	8.353001	0.103963	8.858448	0.366448
2	8.338516	0.094784	8.830649	0.365495
3	4.721753	0.108184	5.297565	0.459198
4	4.671568	0.107004	5.192759	0.436647

Figure 4



The best model with the train-test split date just prior to Trump's presidency had an error of 1.73 (MSE), and more accurately predicted the testing data. Figure 5 displays the actual predicted values alongside the values for perfect one-to-one predictions. It should be mentioned that the axis values for this graph are more limited in range in comparison with Figure 4. From this graph, we see that the model performed with error, but still quite well. I therefore used this model to forecast the import price index for future dates. The results of this forecast are in Figure 6. The forecast predicts a dip in the import price index at the end of 2025, an increase at the beginning of 2026, and then an overall decrease by the end of 2026. Given that these are predictions for the future, we have no way to confirm the accuracy of these predictions. Indeed, it would be very interesting to validate these predictions with observable data in the coming year. I look forward to doing so.

Figure 5

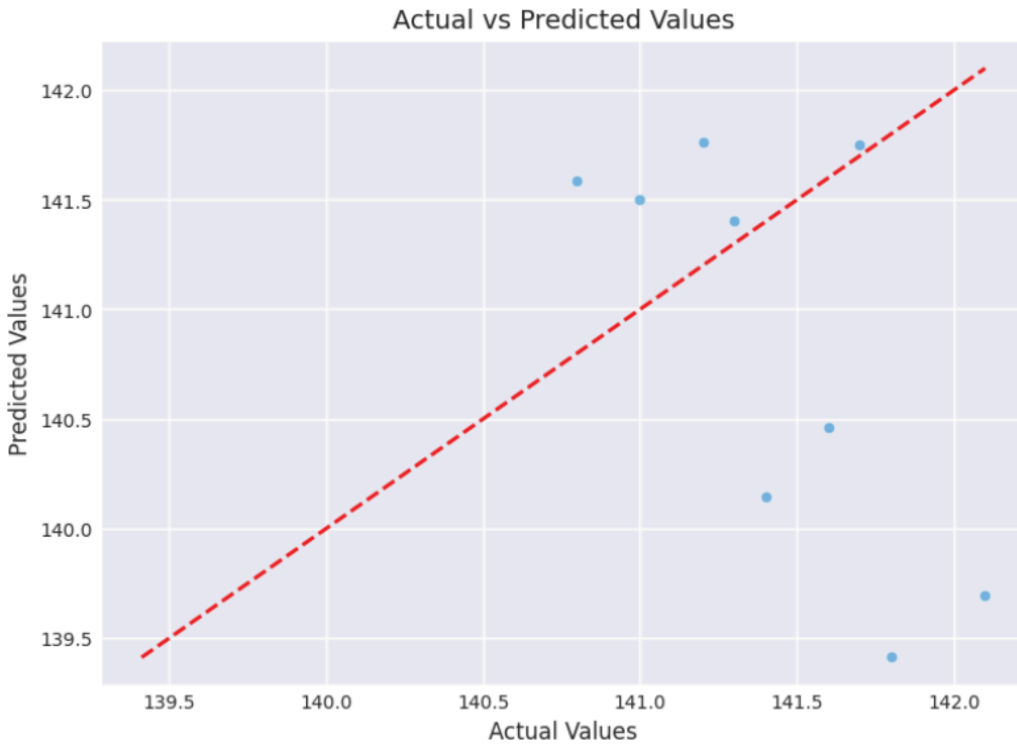
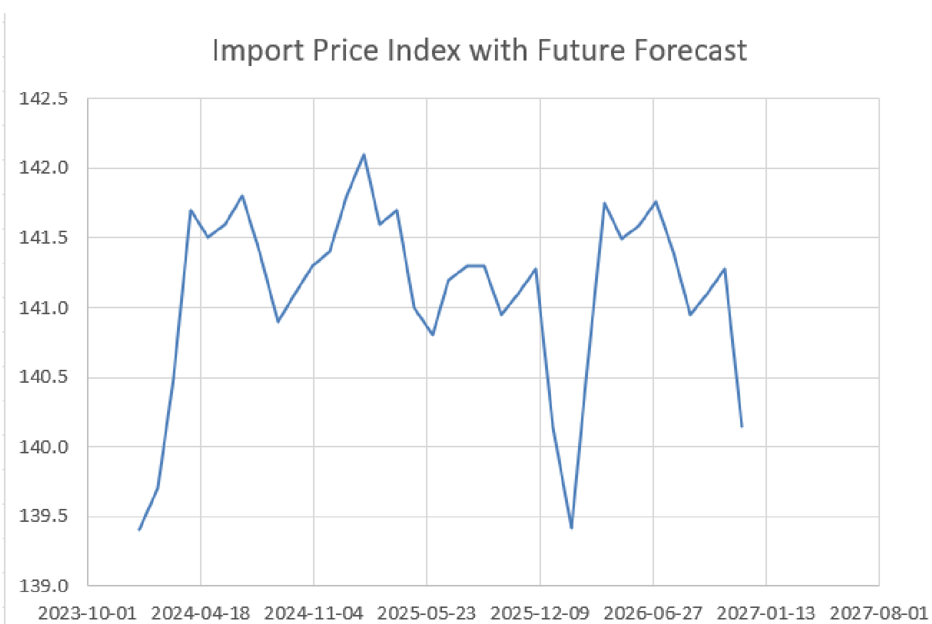


Figure 6



The results from the multivariable OLS regression of log(price) are shown in Figure 7. The R-squared of the regression was only 0.006, which signifies that the variables used in the regression cannot properly explain the change in domestic prices, as measured by yearly inflation. In congruity with what we would expect from the correlation matrix, the coefficients on the features are (weakly) negative, except for the share of exports in 2021. I believe this signifies that the data used simply wasn't robust enough to estimate the causal effect on log(prices). This is also supported by the p-values, which are greater than 0.05, meaning that the coefficients cannot be interpreted as statistically significant and the features cannot be assumed to have any significant causal effect on log(price).

Figure 7

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.006
Model:                            OLS   Adj. R-squared:                 -0.027
Method:                 Least Squares   F-statistic:                    0.4247
Date:                Thu, 18 Dec 2025   Prob (F-statistic):              0.831
Time:                        01:22:19   Log-Likelihood:                 102.15
No. Observations:                 154   AIC:                            -192.3
Df Residuals:                     148   BIC:                            -174.1
Df Model:                           5
Covariance Type:                  HC3
=========================================================================================
                          coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
Intercept              -0.0301      0.018     -1.650      0.099      -0.066       0.006
Ave_Tariff_Increase    -0.0618      0.104     -0.592      0.554      -0.266       0.143
Imports_Importance     -0.0333      0.508     -0.066      0.948      -1.029       0.962
Share_Exports_2022     -0.0540      0.069     -0.780      0.435      -0.190       0.082
Share_Exports_2021      0.0677      0.088      0.773      0.440      -0.104       0.240
Share_Exports_2020     -0.0170      0.094     -0.182      0.856      -0.201       0.167
==============================================================================
Omnibus:                      246.425   Durbin-Watson:                   1.862
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            30075.847
Skew:                           6.933   Prob(JB):                         0.00
Kurtosis:                      70.044   Cond. No.                         144.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Finally, we come to examine the simple full pass-through model. By summing the percent increase attributed to each country, we get a total of 1.61%. This means that the current implemented tariffs are estimated to raise domestic prices by 1.61%. This prediction will also be tested as observable data becomes available in 2026.

From the XGBoost time series regression on the import price index, we can expect the price of imported goods to slightly rise in 2026, relative to the observed values at the end of 2025, but to be relatively stable. This is fairly congruent with the 1.61% increase in domestic prices calculated with the full pass-through.

To conclude with a broader perspective, I present a forecast for the import price index with historical data from 1988 to 2025 and predictions for 2026. While it is true that the 2025 tariffs will temporarily disrupt US trade relations, this does not mean that the economy will be in turmoil forever. Still, for those of us residing in the US, we should prepare for the cost of everything becoming slightly more expensive.